

## 1. General Description

The EM74HC138 and EM74HCT138 decode three binary weighted address inputs (A0, A1 and A2) to eight mutually exclusive outputs ( $\bar{Y}0$  to  $\bar{Y}7$ ). The device features three enable inputs ( $\bar{E}1$ ,  $\bar{E}2$  and E3). Every output will be HIGH unless  $\bar{E}1$  and  $\bar{E}2$  are LOW and E3 is HIGH. This multiple enable function allows easy parallel expansion to a 1-of-32 (5 to 32 lines) decoder with just four '138 ICs and one inverter. The '138 can be used as an eight output demultiplexer by using one of the active LOW enable inputs as the data input and the remaining enable inputs as strobes. Inputs include clamp diodes. This enables the use of current limiting resistors to interface inputs to voltages in excess of  $V_{CC}$ .

## 2. Features and Benefits

- Wide operating voltage 2.0 V to 6.0 V
- High noise immunity
- CMOS low power dissipation
- Latch-up performance exceeds 250 mA
- Demultiplexing capability
- Multiple input enable for easy expansion
- Ideal for memory chip select decoding
- Active LOW mutually exclusive outputs
- Input levels:
  - For EM74HC138: CMOS level
  - For EM74HCT138: TTL level
- ESD protection:
  - HBM: ANSI/ESDA/JEDEC JS-001 class 2 exceeds 3500 V
  - CDM: ANSI/ESDA/JEDEC JS-002 class C3 exceeds 2000 V
- Multiple package options
- Specified from -40 °C to +85 °C and from -40 °C to +125 °C

## EM74HC138; EM74HCT138

3-to-8 line decoder/demultiplexer; inverting

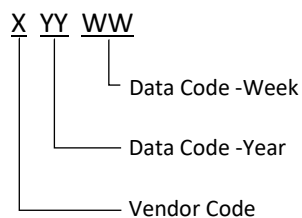
### 3. Ordering Information

**Table 1. Ordering information**

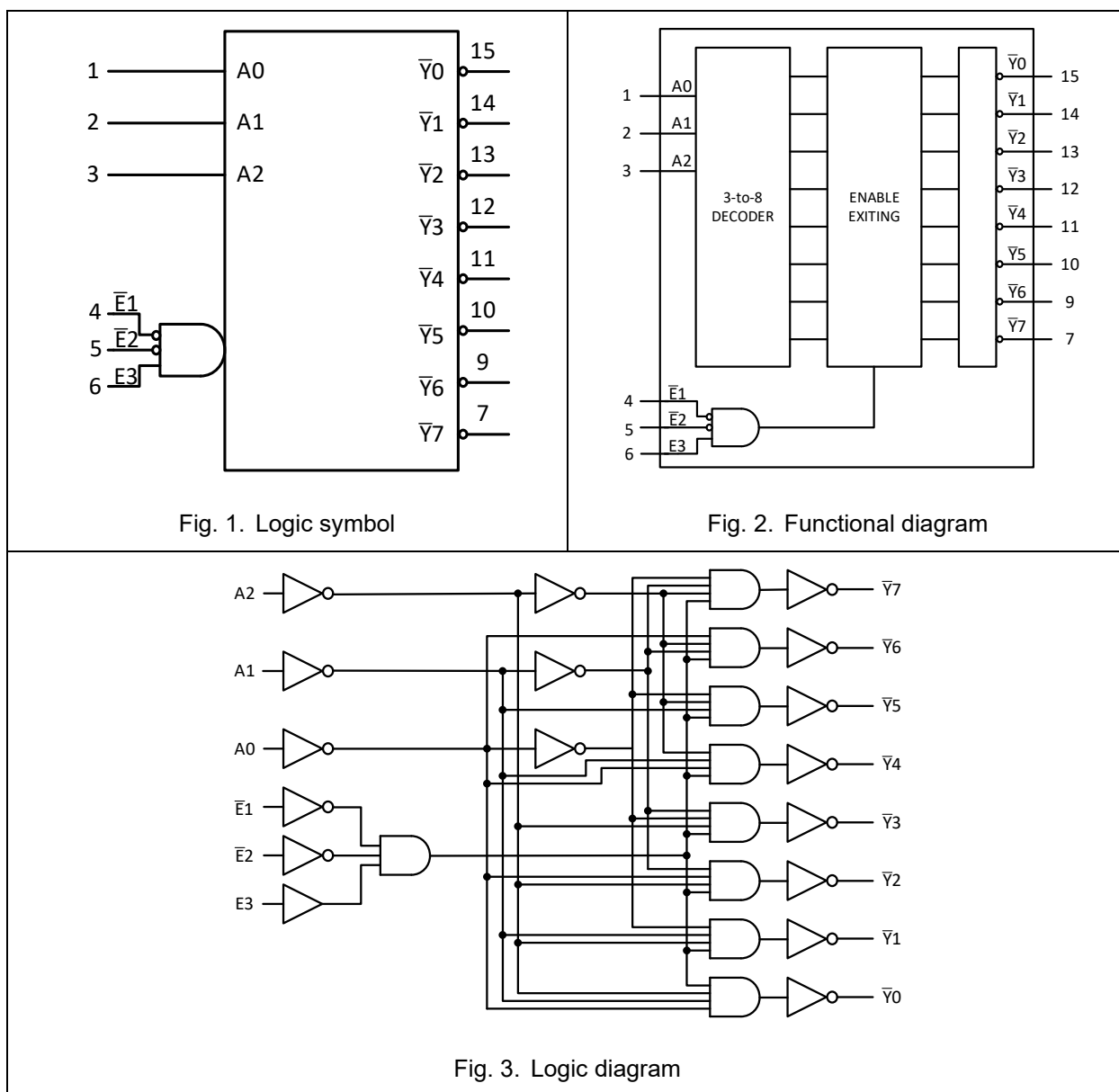
| Type number  | Topside marking | Package   |                                                                           |          |
|--------------|-----------------|-----------|---------------------------------------------------------------------------|----------|
|              |                 | Name      | Description                                                               | Quantity |
| EM74HC138D   | HC138<br>XYYWW  | SOP-16L   | plastic small outline package; 16 leads;<br>body width 3.9 mm             | 3000     |
| EM74HCT138D  | HCT138<br>XYYWW |           |                                                                           |          |
| EM74HC138PW  | HC138<br>XYYWW  | TSSOP-16L | plastic thin shrink small outline package;<br>16 leads; body width 4.4 mm | 3000     |
| EM74HCT138PW | HCT138<br>XYYWW |           |                                                                           |          |

#### MARKING INFORMATION

NOTE: XYYWW = Vendor Code and Data Code.



## 4. Functional diagram



## 5. Pinning Information

### 5.1. Pinning

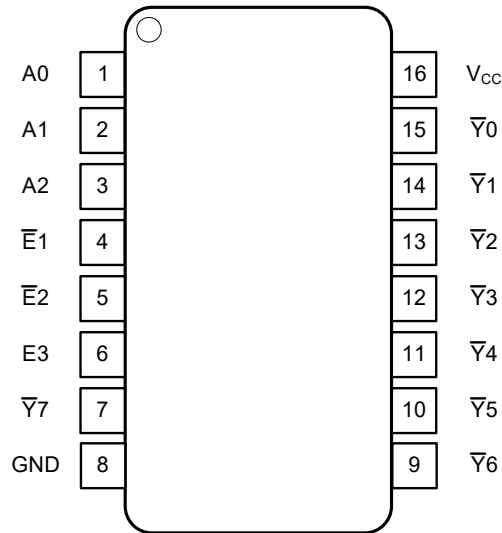


Fig. 4. Top view pin configuration SOP-20L and TSSOP-20L

### 5.2. Pin description

Table 2. Pin description

| Symbol                                                                           | Pin                          | Description                |
|----------------------------------------------------------------------------------|------------------------------|----------------------------|
| A0, A1, A2                                                                       | 1, 2, 3                      | address input              |
| $\bar{E}1, \bar{E}2$                                                             | 4, 5                         | enable input (active LOW)  |
| E3                                                                               | 6                            | enable input (active HIGH) |
| $\bar{Y}0, \bar{Y}1, \bar{Y}2, \bar{Y}3, \bar{Y}4, \bar{Y}5, \bar{Y}6, \bar{Y}7$ | 15, 14, 13, 12, 11, 10, 9, 7 | output                     |
| GND                                                                              | 8                            | ground (0 V)               |
| V <sub>CC</sub>                                                                  | 16                           | supply voltage             |

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# 6.Functional Description

**Table 3. Function table**

H = HIGH voltage level; L = LOW voltage level; X = don't care.

| Control    |            |    | Input |    |    | Output     |            |            |            |            |            |            |            |
|------------|------------|----|-------|----|----|------------|------------|------------|------------|------------|------------|------------|------------|
| $\bar{E}1$ | $\bar{E}2$ | E3 | A2    | A1 | A0 | $\bar{Y}7$ | $\bar{Y}6$ | $\bar{Y}5$ | $\bar{Y}4$ | $\bar{Y}3$ | $\bar{Y}2$ | $\bar{Y}1$ | $\bar{Y}0$ |
| H          | X          | X  | X     | X  | X  | H          | H          | H          | H          | H          | H          | H          | H          |
| X          | H          | X  |       |    |    |            |            |            |            |            |            |            |            |
| X          | X          | L  |       |    |    |            |            |            |            |            |            |            |            |
| L          | L          | H  | L     | L  | L  | H          | H          | H          | H          | H          | H          | H          | L          |
|            |            |    | L     | L  | H  | H          | H          | H          | H          | H          | H          | L          | H          |
|            |            |    | L     | H  | L  | H          | H          | H          | H          | H          | L          | H          | H          |
|            |            |    | L     | H  | H  | H          | H          | H          | H          | L          | H          | H          | H          |
|            |            |    | H     | L  | L  | H          | H          | H          | L          | H          | H          | H          | H          |
|            |            |    | H     | L  | H  | H          | H          | L          | H          | H          | H          | H          | H          |
|            |            |    | H     | H  | L  | H          | L          | H          | H          | H          | H          | H          | H          |
|            |            |    | H     | H  | H  | L          | H          | H          | H          | H          | H          | H          | H          |

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# 7. Absolute Maximum Ratings

Stresses exceeding the absolute maximum ratings may damage the device. The device may not function or be operable above the recommended operating conditions and stressing the parts to these levels is not recommended. In addition, extended exposure to stresses above the recommended operating conditions may affect device reliability. The absolute maximum ratings are stress ratings only.

**Table 4. Absolute Maximum Ratings**

In accordance with the Absolute Maximum Rating System (IEC 60134). Voltages are referenced to GND.

| Symbol    | Parameter               | Conditions                                                                | Min  | Max      | Unit               |
|-----------|-------------------------|---------------------------------------------------------------------------|------|----------|--------------------|
| $V_{CC}$  | supply voltage          |                                                                           | -0.5 | 7        | V                  |
| $I_{IK}$  | input clamping current  | $V_I < -0.5\text{ V}$ or $V_I > V_{CC} + 0.5\text{ V}$                    |      | $\pm 20$ | mA                 |
| $I_{OK}$  | output clamping current | $V_O < -0.5\text{ V}$ or $V_O > V_{CC} + 0.5\text{ V}$                    |      | $\pm 20$ | mA                 |
| $I_O$     | output current          | $-0.5\text{ V} < V_O < V_{CC} + 0.5\text{ V}$                             |      | $\pm 25$ | mA                 |
| $I_{CC}$  | supply current          |                                                                           |      | 50       | mA                 |
| $I_{GND}$ | ground current          |                                                                           | -50  |          | mA                 |
| $P_{tot}$ | total power dissipation | $T_{amb} = -40\text{ }^{\circ}\text{C}$ to $+125\text{ }^{\circ}\text{C}$ |      | 500      | mW                 |
| $T_{stg}$ | storage temperature     |                                                                           | -65  | +150     | $^{\circ}\text{C}$ |

# 8. Recommended Operating Conditions

The Recommended Operating Conditions table defines the conditions for actual device operation. Recommended operating conditions are specified to ensure optimal performance to the datasheet specifications. EnergyMath does not recommend exceeding them or designing to Absolute Maximum Ratings.

**Table 5. Recommended Operating Conditions**

| Symbol              | Parameter                           | Conditions              | EM74HC138 |      |          | EM74HCT138 |      |          | Unit               |
|---------------------|-------------------------------------|-------------------------|-----------|------|----------|------------|------|----------|--------------------|
|                     |                                     |                         | Min       | Min  | Typ      | Min        | Min  | Typ      |                    |
| $V_{CC}$            | supply voltage                      |                         | 2.0       | 5.0  | 6.0      | 4.5        | 5.0  | 5.5      | V                  |
| $V_I$               | input voltage                       |                         | 0         |      | $V_{CC}$ | 0          |      | $V_{CC}$ | V                  |
| $V_O$               | output voltage                      |                         | 0         |      | $V_{CC}$ | 0          |      | $V_{CC}$ | V                  |
| $T_{amb}$           | ambient temperature                 |                         | -40       | +25  | +125     | -40        | +25  | +125     | $^{\circ}\text{C}$ |
| $\Delta t/\Delta V$ | input transition rise and fall rate | $V_{CC} = 2.0\text{ V}$ |           |      | 625      |            |      |          | ns/V               |
|                     |                                     | $V_{CC} = 4.5\text{ V}$ |           | 1.67 | 139      |            | 1.67 | 139      | ns/V               |
|                     |                                     | $V_{CC} = 6.0\text{ V}$ |           |      | 83       |            |      |          | ns/V               |

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3-to-8 line decoder/demultiplexer; inverting

# 9.Static Characteristics

**Table 6. Static characteristics**

At recommended operating conditions. Voltages are referenced to GND (ground = 0 V). Typical values measured at  $T_{amb} = 25^{\circ}\text{C}$  (unless otherwise noted).

| Symbol          | Parameter                 | Conditions                                                                                  | -40 °C to +85 °C |     |      | -40 °C to +125 °C |      | Unit |
|-----------------|---------------------------|---------------------------------------------------------------------------------------------|------------------|-----|------|-------------------|------|------|
|                 |                           |                                                                                             | Min              | Typ | Max  | Min               | Max  |      |
| EM74HC138       |                           |                                                                                             |                  |     |      |                   |      |      |
| V <sub>IH</sub> | HIGH-level input voltage  | V <sub>CC</sub> = 2.0 V                                                                     | 1.5              |     |      | 1.5               |      | V    |
|                 |                           | V <sub>CC</sub> = 4.5 V                                                                     | 3.15             |     |      | 3.15              |      | V    |
|                 |                           | V <sub>CC</sub> = 6.0 V                                                                     | 4.2              |     |      | 4.2               |      | V    |
| V <sub>IL</sub> | LOW-level input voltage   | V <sub>CC</sub> = 2.0 V                                                                     |                  |     | 0.5  |                   | 0.5  | V    |
|                 |                           | V <sub>CC</sub> = 4.5 V                                                                     |                  |     | 1.35 |                   | 1.35 | V    |
|                 |                           | V <sub>CC</sub> = 6.0 V                                                                     |                  |     | 1.8  |                   | 1.8  | V    |
| V <sub>OH</sub> | HIGH-level output voltage | V <sub>I</sub> = V <sub>IH</sub> or V <sub>IL</sub>                                         |                  |     |      |                   |      |      |
|                 |                           | I <sub>O</sub> = -20 μA; V <sub>CC</sub> = 2.0 V                                            | 1.9              |     |      | 1.9               |      | V    |
|                 |                           | I <sub>O</sub> = -20 μA; V <sub>CC</sub> = 4.5 V                                            | 4.4              |     |      | 4.4               |      | V    |
|                 |                           | I <sub>O</sub> = -20 μA; V <sub>CC</sub> = 6.0 V                                            | 5.9              |     |      | 5.9               |      | V    |
|                 |                           | I <sub>O</sub> = -4.0 mA; V <sub>CC</sub> = 4.5 V                                           | 3.84             |     |      | 3.7               |      | V    |
|                 |                           | I <sub>O</sub> = -5.2 mA; V <sub>CC</sub> = 6.0 V                                           | 5.34             |     |      | 5.2               |      | V    |
| V <sub>OL</sub> | LOW-level output voltage  | V <sub>I</sub> = V <sub>IH</sub> or V <sub>IL</sub>                                         |                  |     |      |                   |      |      |
|                 |                           | I <sub>O</sub> = 20 μA; V <sub>CC</sub> = 2.0 V                                             |                  |     | 0.1  |                   | 0.1  | V    |
|                 |                           | I <sub>O</sub> = 20 μA; V <sub>CC</sub> = 4.5 V                                             |                  |     | 0.1  |                   | 0.1  | V    |
|                 |                           | I <sub>O</sub> = 20 μA; V <sub>CC</sub> = 6.0 V                                             |                  |     | 0.1  |                   | 0.1  | V    |
|                 |                           | I <sub>O</sub> = 4.0 mA; V <sub>CC</sub> = 4.5 V                                            |                  |     | 0.33 |                   | 0.4  | V    |
|                 |                           | I <sub>O</sub> = 5.2 mA; V <sub>CC</sub> = 6.0 V                                            |                  |     | 0.33 |                   | 0.4  | V    |
| I <sub>I</sub>  | input leakage current     | V <sub>I</sub> = V <sub>CC</sub> or GND ;<br>V <sub>CC</sub> = 6.0 V                        |                  |     | ±1   |                   | ±1   | μA   |
| I <sub>CC</sub> | supply current            | V <sub>I</sub> = V <sub>CC</sub> or GND ; I <sub>O</sub> = 0 A ;<br>V <sub>CC</sub> = 6.0 V |                  |     | 20   |                   | 40   | μA   |
| C <sub>I</sub>  | input capacitance         |                                                                                             |                  | 4.0 |      |                   |      | pF   |

## EM74HC138; EM74HCT138

3-to-8 line decoder/demultiplexer; inverting

| Symbol           | Parameter                 | Conditions                                                                                                                                               | -40 °C to +85 °C |     |      | -40 °C to +125 °C |     | Unit |
|------------------|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-----|------|-------------------|-----|------|
|                  |                           |                                                                                                                                                          | Min              | Typ | Max  | Min               | Max |      |
| EM74HCT138       |                           |                                                                                                                                                          |                  |     |      |                   |     |      |
| V <sub>IH</sub>  | HIGH-level input voltage  | V <sub>CC</sub> = 4.5 V to 5.5 V                                                                                                                         | 2.0              |     |      | 2.0               |     | V    |
| V <sub>IL</sub>  | LOW-level input voltage   | V <sub>CC</sub> = 4.5 V to 5.5 V                                                                                                                         |                  |     | 0.8  |                   | 0.8 | V    |
| V <sub>OH</sub>  | HIGH-level output voltage | V <sub>I</sub> = V <sub>IH</sub> or V <sub>IL</sub> ;                                                                                                    |                  |     |      |                   |     |      |
|                  |                           | I <sub>O</sub> = -20 μA; V <sub>CC</sub> = 4.5 V                                                                                                         | 4.4              |     |      | 4.4               |     | V    |
|                  |                           | I <sub>O</sub> = -4.0 mA; V <sub>CC</sub> = 4.5 V                                                                                                        | 3.84             |     |      | 3.7               |     | V    |
| V <sub>OL</sub>  | LOW-level output voltage  | V <sub>I</sub> = V <sub>IH</sub> or V <sub>IL</sub> ;                                                                                                    |                  |     |      |                   |     |      |
|                  |                           | I <sub>O</sub> = 20 μA; V <sub>CC</sub> = 4.5 V                                                                                                          |                  |     | 0.1  |                   | 0.1 | V    |
|                  |                           | I <sub>O</sub> = 4.0 mA; V <sub>CC</sub> = 4.5 V                                                                                                         |                  |     | 0.33 |                   | 0.4 | V    |
| I <sub>I</sub>   | input leakage current     | V <sub>I</sub> = V <sub>CC</sub> or GND ;<br>V <sub>CC</sub> = 5.5 V                                                                                     |                  |     | ±1   |                   | ±1  | μA   |
| I <sub>CC</sub>  | supply current            | V <sub>I</sub> = V <sub>CC</sub> or GND; I <sub>O</sub> = 0 A;<br>V <sub>CC</sub> = 5.5 V                                                                |                  |     | 20   |                   | 40  | μA   |
| ΔI <sub>CC</sub> | additional supply current | per pin ; V <sub>I</sub> = V <sub>CC</sub> - 2.1 V; I <sub>O</sub> = 0 A;<br>other inputs at V <sub>CC</sub> or GND;<br>V <sub>CC</sub> = 4.5 V to 5.5 V |                  |     | 400  |                   | 490 | μA   |
| C <sub>I</sub>   | input capacitance         |                                                                                                                                                          |                  | 4.0 |      |                   |     | pF   |

## EM74HC138; EM74HCT138

3-to-8 line decoder/demultiplexer; inverting

# 10. Dynamic Characteristics

**Table 7. Dynamic characteristics**

Voltages are referenced to GND (ground = 0 V); for test circuit see Fig. 7. Typical values measured at  $T_{amb} = 25^{\circ}\text{C}$  (unless otherwise noted).

| Symbol          | Parameter                     | Conditions                                                                         | -40 °C to +85 °C |     |     | -40 °C to +125 °C |     | Unit |
|-----------------|-------------------------------|------------------------------------------------------------------------------------|------------------|-----|-----|-------------------|-----|------|
|                 |                               |                                                                                    | Min              | Typ | Max | Min               | Max |      |
| EM74HC138       |                               |                                                                                    |                  |     |     |                   |     |      |
| t <sub>pd</sub> | propagation delay             | An to $\bar{Y}_n$ ; see Fig. 5 [1]                                                 |                  |     |     |                   |     |      |
|                 |                               | V <sub>CC</sub> = 2.0 V                                                            |                  |     | 120 |                   | 130 | ns   |
|                 |                               | V <sub>CC</sub> = 4.5 V                                                            |                  |     | 37  |                   | 42  | ns   |
|                 |                               | V <sub>CC</sub> = 6.0 V                                                            |                  |     | 30  |                   | 33  | ns   |
|                 |                               | E3 to $\bar{Y}_n$ ; see Fig. 5 [1]                                                 |                  |     |     |                   |     |      |
|                 |                               | V <sub>CC</sub> = 2.0 V                                                            |                  |     | 120 |                   | 130 | ns   |
|                 |                               | V <sub>CC</sub> = 4.5 V                                                            |                  |     | 37  |                   | 42  | ns   |
|                 |                               | V <sub>CC</sub> = 6.0 V                                                            |                  |     | 30  |                   | 33  | ns   |
|                 |                               | $\bar{E}_n$ to $\bar{Y}_n$ ; see Fig. 6 [1]                                        |                  |     |     |                   |     |      |
|                 |                               | V <sub>CC</sub> = 2.0 V                                                            |                  |     | 120 |                   | 130 | ns   |
|                 |                               | V <sub>CC</sub> = 4.5 V                                                            |                  |     | 37  |                   | 42  | ns   |
|                 |                               | V <sub>CC</sub> = 6.0 V                                                            |                  |     | 30  |                   | 33  | ns   |
| t <sub>t</sub>  | transition time               | $\bar{Y}_n$ ; see Fig. 5 and Fig. 6 [2]                                            |                  |     |     |                   |     |      |
|                 |                               | V <sub>CC</sub> = 2.0 V                                                            |                  |     | 9   |                   | 11  | ns   |
|                 |                               | V <sub>CC</sub> = 4.5 V                                                            |                  |     | 6   |                   | 8   | ns   |
|                 |                               | V <sub>CC</sub> = 6.0 V                                                            |                  |     | 4   |                   | 5   | ns   |
| C <sub>PD</sub> | power dissipation capacitance | C <sub>L</sub> = 50pF ; f = 1 MHz ;<br>V <sub>I</sub> = GND to V <sub>CC</sub> [3] |                  | 20  |     |                   |     | pF   |

## EM74HC138; EM74HCT138

3-to-8 line decoder/demultiplexer; inverting

| Symbol          | Parameter                     | Conditions                                                                              | -40 °C to +85 °C |     |     | -40 °C to +125 °C |     | Unit |
|-----------------|-------------------------------|-----------------------------------------------------------------------------------------|------------------|-----|-----|-------------------|-----|------|
|                 |                               |                                                                                         | Min              | Typ | Max | Min               | Max |      |
| EM74HCT138      |                               |                                                                                         |                  |     |     |                   |     |      |
| t <sub>pd</sub> | propagation delay             | An to $\bar{Y}_n$ ; V <sub>CC</sub> = 4.5V; see Fig. 5 [1]                              |                  |     | 37  |                   | 42  | ns   |
|                 |                               | E3 to $\bar{Y}_n$ ; V <sub>CC</sub> = 4.5V; see Fig. 5 [1]                              |                  |     | 37  |                   | 42  | ns   |
|                 |                               | $\bar{E}_n$ to $\bar{Y}_n$ ; V <sub>CC</sub> = 4.5V see Fig. 6 [1]                      |                  |     | 37  |                   | 42  | ns   |
| t <sub>t</sub>  | transition time               | $\bar{Y}_n$ ; V <sub>CC</sub> =4.5V; see Fig. 5 and Fig. 6; [2]                         |                  |     | 6   |                   | 8   | ns   |
| C <sub>PD</sub> | power dissipation capacitance | C <sub>L</sub> = 50pF ; f = 1 MHz ; V <sub>I</sub> = GND to V <sub>CC</sub> – 1.5 V [3] |                  | 20  |     |                   |     | pF   |

[1]  $t_{pd}$  is the same as  $t_{PHL}$  and  $t_{PLH}$ .

[2]  $t_t$  is the same as  $t_{THL}$  and  $t_{TLH}$ .

[3]  $C_{PD}$  is used to determine the dynamic power dissipation ( $P_D$  in  $\mu W$ ).

$P_D = C_{PD} \times V_{CC}^2 \times f_i \times N + \Sigma(C_L \times V_{CC}^2 \times f_o)$  where:

$f_i$  = input frequency in MHz;

$f_o$  = output frequency in MHz;

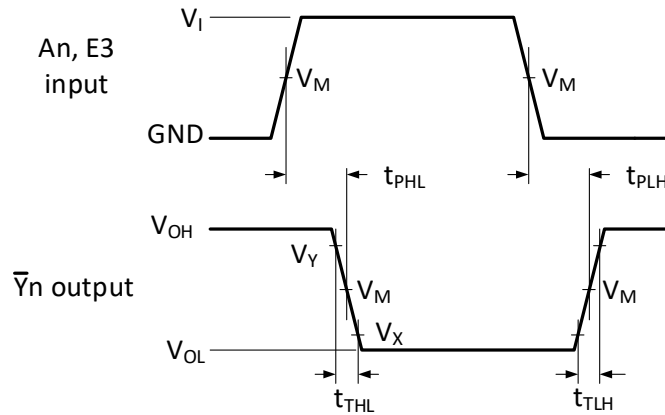
$C_L$  = output load capacitance in pF;

$V_{CC}$  = supply voltage in V;

$N$  = number of inputs switching;

$\Sigma(C_L \times V_{CC}^2 \times f_o)$  = sum of outputs.

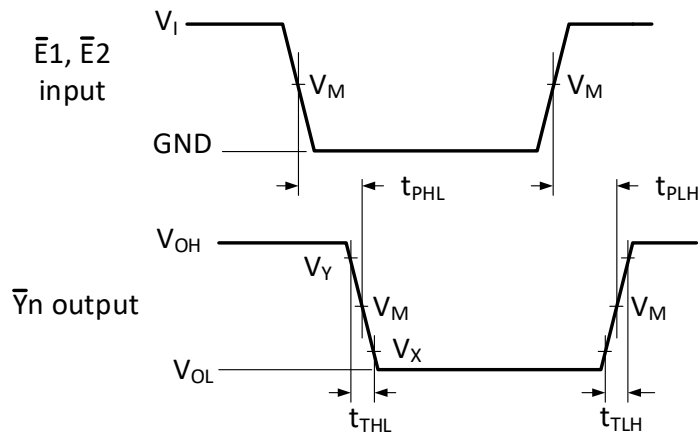
## 10.1. Waveforms and test circuit



Measurement points are given in Table 8.

$V_{OL}$  and  $V_{OH}$  are typical output voltage levels that occur with the output load.

Fig. 5. Propagation delay input (An) and enable input (E3) to output ( $\bar{Y}_n$ ) and transition time output ( $\bar{Y}_n$ )



Measurement points are given in Table 8.

$V_{OL}$  and  $V_{OH}$  are typical output voltage levels that occur with the output load.

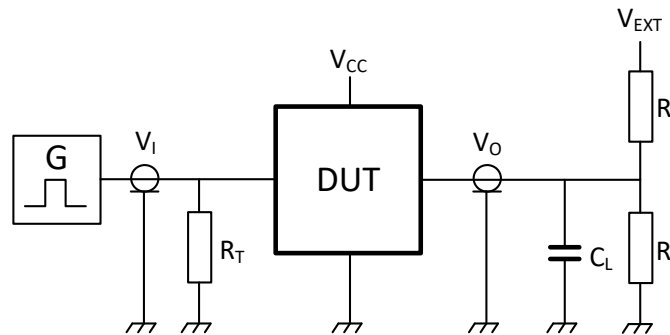
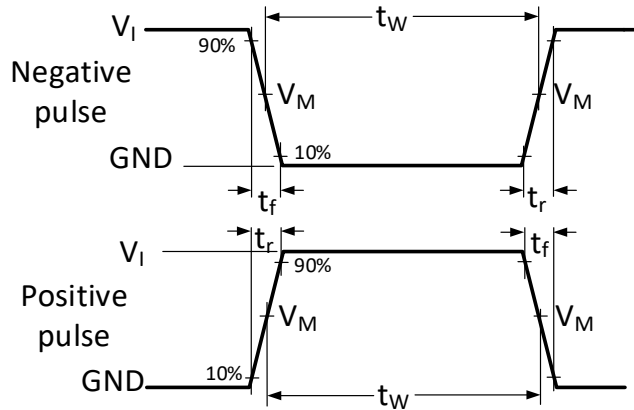
Fig. 6. Propagation delay enable input ( $\bar{E}_n$ ) to output ( $\bar{Y}_n$ ) and transition time output ( $\bar{Y}_n$ )

**Table 8. Measurement points**

| Type       | Input       | Output      |             |             |
|------------|-------------|-------------|-------------|-------------|
|            | $V_M$       | $V_M$       | $V_X$       | $V_Y$       |
| EM74HC138  | $0.5V_{CC}$ | $0.5V_{CC}$ | $0.1V_{CC}$ | $0.9V_{CC}$ |
| EM74HCT138 | 1.3 V       | 1.3 V       | $0.1V_{CC}$ | $0.9V_{CC}$ |

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Test data is given in Table 9.

Definitions for test circuit:

$R_L$  = Load resistance.

$C_L$  = Load capacitance including jig and probe capacitance.

$R_T$  = Termination resistance should be equal to the output impedance  $Z_o$  of the pulse generator.

$V_{EXT}$  = External voltage for measuring switching times.

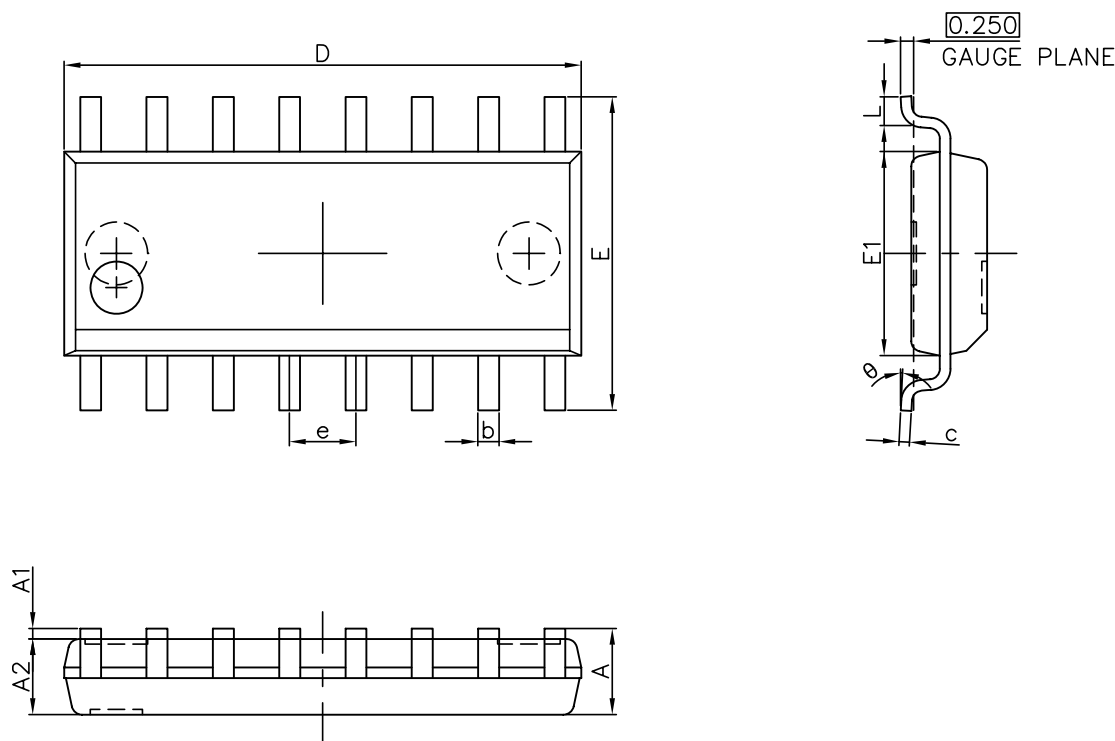
Fig. 7. Test circuit for measuring switching times

Table 9. Test data

| Type       | Input    |               | Load  |              | $V_{EXT}$          |                    |                    |
|------------|----------|---------------|-------|--------------|--------------------|--------------------|--------------------|
|            | $V_I$    | $t_r = t_f$   | $C_L$ | $R_L$        | $t_{PLH}, t_{PHL}$ | $t_{PZH}, t_{PHZ}$ | $t_{PZL}, t_{PLZ}$ |
| EM74HC138  | $V_{CC}$ | $\leq 2.5$ ns | 50 pF | 500 $\Omega$ | open               | GND                | $2V_{CC}$          |
| EM74HCT138 | 3 V      | $\leq 2.5$ ns | 50 pF | 500 $\Omega$ | open               | GND                | $2V_{CC}$          |

## 11. Package Outline

### SOP-16L

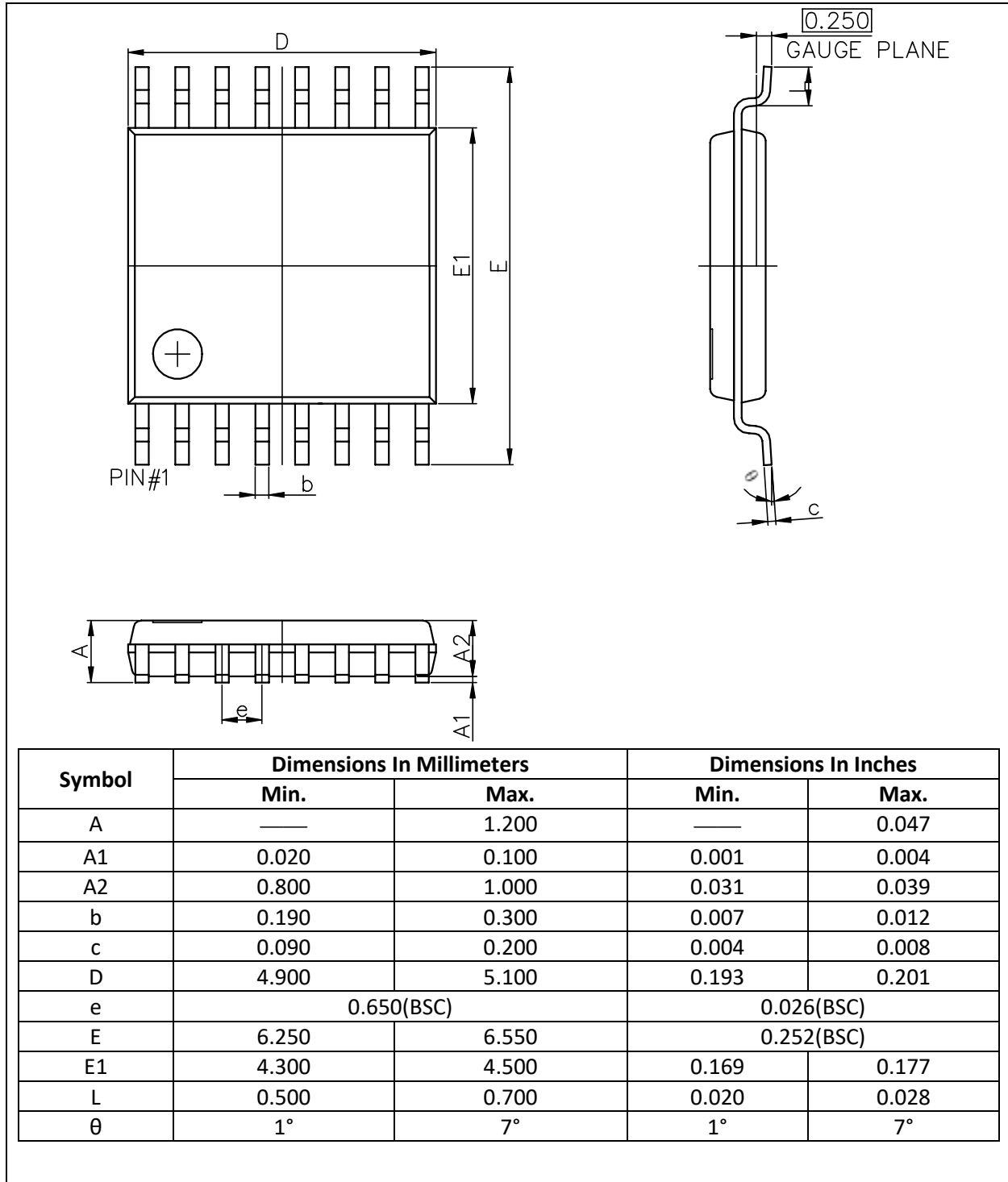


| Symbol | Dimensions In Millimeters |        | Dimensions In Inches |       |
|--------|---------------------------|--------|----------------------|-------|
|        | Min.                      | Max.   | Min.                 | Max.  |
| A      | —                         | 1.750  | —                    | 0.069 |
| A1     | 0.150                     | 0.250  | 0.006                | 0.010 |
| A2     | 1.400                     | 1.500  | 0.055                | 0.059 |
| b      | 0.330                     | 0.510  | 0.013                | 0.020 |
| c      | 0.170                     | 0.250  | 0.007                | 0.010 |
| D      | 9.800                     | 10.000 | 0.386                | 0.394 |
| e      | 1.270(BSC)                |        | 0.050(BSC)           |       |
| E      | 5.900                     | 6.100  | 0.232                | 0.240 |
| E1     | 3.800                     | 4.000  | 0.150                | 0.157 |
| L      | 0.400                     | 1.270  | 0.016                | 0.050 |
| θ      | 0°                        | 8°     | 0°                   | 8°    |

# EM74HC138; EM74HCT138

3-to-8 line decoder/demultiplexer; inverting

## TSSOP-16L



## 12. Abbreviations

**Table 10. Abbreviations**

| Acronym | Description                             |
|---------|-----------------------------------------|
| CMOS    | Complementary Metal-Oxide Semiconductor |
| DUT     | Device Under Test                       |
| ESD     | ElectroStatic Discharge                 |
| HBM     | Human Body Model                        |
| CDM     | Charged Device Model                    |
| TTL     | Transistor-Transistor Logic             |

## 13. Revision History

**Table 11. Revision history**

| Document ID            | Release Date | Data sheet status | Change notice | Supersedes |
|------------------------|--------------|-------------------|---------------|------------|
| EM74HC_HCT138 Rev. 1.0 | Sep 24, 2024 | Product datasheet |               |            |